

Blessing Mbalaka

Projects Worked on:

August, September, October, November

1. STEM-APP

The application was built from scratch. The front-end was provided the key features provided in the application included:

1. Video conferencing interface—an in app zoom fallback for tutor sessions
2. Back-end logic for APIs for the gamification feature.
3. Custom Game maker in the admin
4. Dashboard analytics was created from google analytics, and from the in-app analytics
5. Chatbot was built from scratch, and adapted for retrieval augmented generation for document only response.
6. The Role based logic, login, role based authentication
7. Forum and image upload functionality.
8. Gamification tokens and animations.

Status Completed. Latest code available for deployment. Version 1 already deployed.

2. CHIETA LMS

70% of the code was done by me.

1. I created the document parsing algorithm that worked to extract images, questions and to allow for structured storage. The logic took time to perfect because it required that their document was preserved exactly to comply with QCTO.
2. I created the randomization logic and the exam writing feature for the student.
3. Created the Status based role-based viewing logic.
4. Did the initial project merge of the code and apps.

Status-completed—Sent for Deployment.

5. Curriculum Chatbot

Key features:

1. The RAG capabilities and engine for searching curriculums.

Status completed.

4. Webscrapping

The webscrapping was done in phases:

- a) The initial phase the target was 10 schools per continent, which were the top 10 from the QS world rankings. Visualisations and data-sets provided.
- b) The outcome yielded promising results with a success rate that provided the initial data-fields.

— After consulting with the Tatenda, I was advised that the data was not sufficient.
 — A month elapsed and I was stold to revert back to the initial code and stratify the outcome.

— I mentioned my grievances which were that the African business schools were limited in supply. Each time i mention that there is no list of for example, 1000 schools my grievances get acknowledged. I checked the business school accreditation bodies and they all had a eurocentric bias with Africa accouting for less than 100 in their representation.

— I provided all the African business school data via email to for all the ones found. Some manually done.

So I decided to finish and provide what i can hence the outcome below:

1. My lead was sent **1800** business schools from Europe, America and approximately **50** from Africa—Due to the challenge of finding a curated list.
2. The African list was sent on the 8th November. The reason for the list is because the African business school directories do not represent African schools.

The challenge:

3. The strategy for scrapping these schools requires a multi-site adaptive scrapper which essentially means creating a unique scrapper for all these schools that is cognizant of the language variance and adaptive to the various site structures, naming conventions.
4. I used a device that was not suited to the job which took days with lack-luster results.

The response to the challenge:

1. Being cognizant of this challenge, I devised a scrapper that went to extract raw site text from the scrappers and it worked by downloading the entire site, so that we can bypass the scrapping restrictions.
2. The scrapper had success with the top 10 schools.
3. I called the African business school branch in Sandton to find out if they had resources i could use for Africa, and they said they only had the 7 accredited.
4. I called the AACSB and tried to get access to their API and the email got no response.
5. I consulted with experts such as Chris and they advised me on the need for finding an API.
6. I requested access to Google's SerAPI but still have not heard back because i wanted to explore if it can be useful to help achieve the projects deliverables in shorter time. Since all data is indexed on google, it would be easier than building individual website scrappers for all African schools when found or dealing with the poolof 1800.
7. I tried it and so results and sent them 3 months a go, but i used a free version and saw promising results, it worked well to gather documents from the targeted universities.

The outcome:

1. Sent 1800 business schools to the email to all project stakeholders. Inclusive of findings from Europe and America.
 2. Sent the limited African Business schools.
 3. Sent already published business school data from America and Europe from 2025.
 4. Another deliverable was building a system, I built it to use-live webscrapping from curated sources to scrape but the speed was something too intense for my machine.
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Publications:

1. The Digitalization proposal draft was sent—Today I see revisions are required to realign.
2. Gamification—Draft complete, but Methodology needed to be revised to use Zandile's expertise, and the journal draft is being realigned to match that.